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processes, which includes the following examples.

1. A waterfall runs downhill, but never uphill, spontaneously.

2. A lump of sugar dissolves spontaneously in a cup of coffee, but never reappears in its original form spontaneously.

3. Heat flows from hotter object to a colder one, but never flows from colder to hotter object spontaneously.

4. The expansion of a gas into an evacuated bulb is a spontaneous process, the reverse process that is gathering of all molecules into one bulb is not. spontaneous.

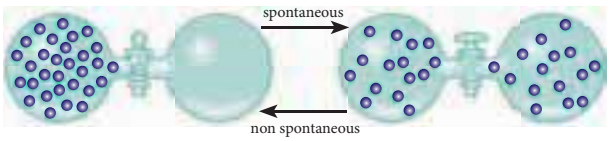


Figure 7.9 Spontaneous process illustration

These examples show that the processes that occur spontaneously in one direction, cannot take place in opposite direction spontaneously.

Similarly a large number of exothermic reactions are spontaneous. An example is combustion of methane.

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2\,H_2O$$

$$\Delta H^o = -890.4 \text{ kJ mol}^{-1}$$

Another example is acid-base neutralization reaction:

$$H^+ + OH^- \longrightarrow H_2O$$

$$\Delta H^o = -57.32 \text{ kJ mol}^{-1}$$

However, some endothermic processes are also spontaneous. For example ammonium nitrate dissolves in water spontaneously though this dissolution is endothermic.

$$NH_4NO_3 \xrightarrow{H_2O} NH_4^+ + NO_3^-$$

$$\Delta H^o = +25 \text{ kJ mol}^{-1}$$

From the above examples we can come to the conclusion that exothermicity favors the spontaneity but does not guarantee it. We cannot decide whether or not a chemical reaction will occur spontaneously solely on the basis of energy changes in the system. We know from second law of thermodynamics that in a spontaneous process, the entropy increases. But not all the processes which are accompanied by an increase in entropy are spontaneous. In order to predict the spontaneity of a reaction, we need some other thermodynamic function.

The second law of thermodynamics introduces another thermodynamic function called Gibbs free energy which finds useful in predicting the spontaneity of a reaction. The Gibbs free energy (G) was developed in the 1870's by Josiah Willard Gibbs. He originally termed this energy as the "available energy" to do work in a system. This quantity is the energy associated with a chemical reaction that can be used to do work.

Gibbs free energy is defined as below

$$G = H - TS \text{ ---------- (7.35)}$$

Gibbs free energy (G) is an extensive property and it is a single valued